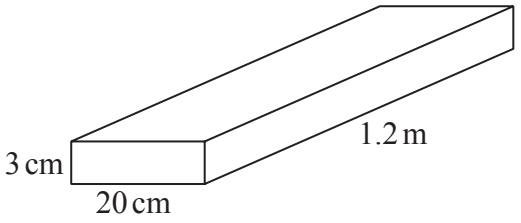
12 The diagram shows a piece of wood in the shape of a cuboid.



The piece of wood is 3 cm by 20 cm by 1.2 m.
The mass of the piece of wood is 8 kg.

The piece of wood will float in sea water if the density of the wood is less than the density of the sea water.

In a large pool, 1 litre of sea water has a mass of 1030 g.

Will the piece of wood float in this pool?
You must show how you get your answer.

1.2 m = 120 cm; 8 kg = 8000 g.

Wood volume = 3 × 20 × 120 = 7200 cm^3.

1 mark
Volume in cm^3

Wood density = 8000 ÷ 7200 = 1.1111... g/cm^3.

1 mark
Mass ÷ volume

Sea water: 1 litre = 1000 cm^3.

Density = 1030 ÷ 1000 = 1.03 g/cm^3.

1 mark
Both densities in same units

No: 1.1111... > 1.03, so the wood is denser.

1 mark
No; correct comparison

(Total for Question 12 is 4 marks)